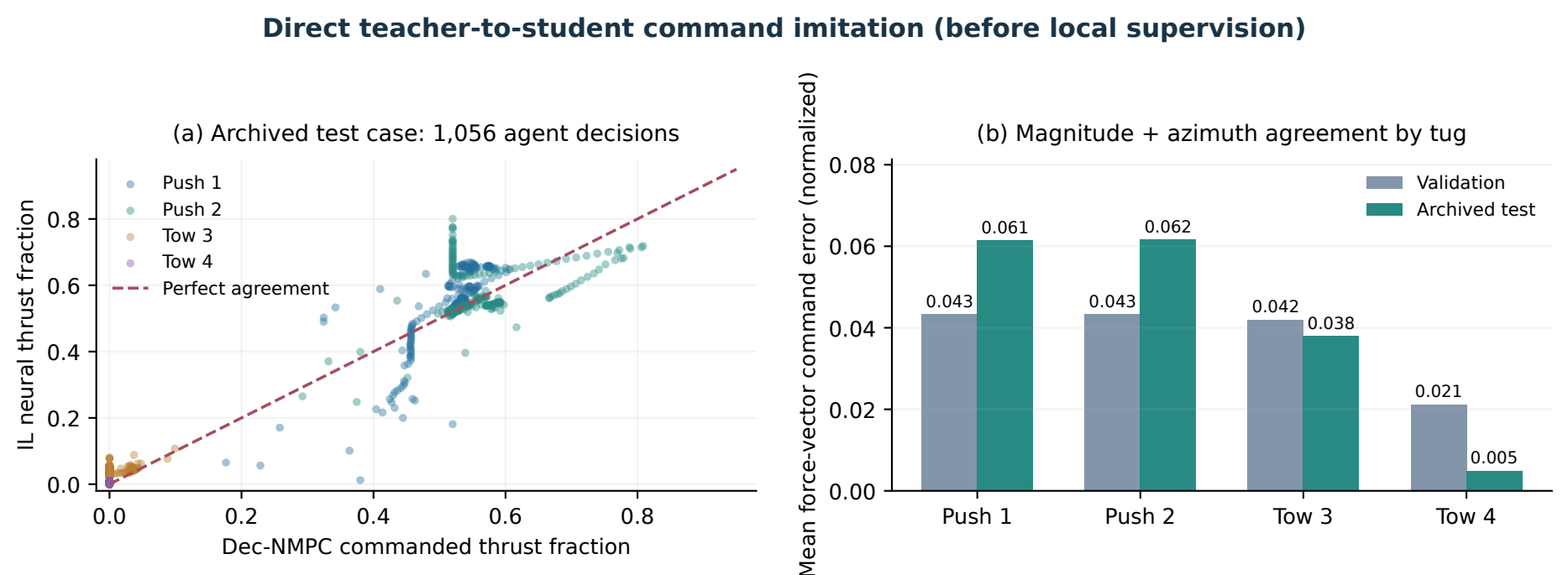
Direct teacher-to-student command imitation (before local supervision)
(a) Archived test case: 1,056 agent decisions
Push 1
Push 2
Tow 3
Tow 4
Perfect agreement
IL neural thrust fraction
Dec-NMPC commanded thrust fraction
0.0
0.2
0.4
0.6
0.8
(b) Magnitude + azimuth agreement by tug
Validation
Archived test
Mean force-vector command error (normalized)
0.00
0.02
0.04
0.06
0.08
0.043
0.061
0.043
0.062
0.042
0.038
0.021
0.005
Push 1
Push 2
Tow 3
Tow 4